

CHARACTER移動控制

坦克車移動方法

- 座標位置的改變
 - 座標移動 `transform.Translate();`
 - 角度旋轉 `transform.Rotate();`
- API Reference
- <https://docs.unity3d.com/ScriptReference/Transform.Translate.html>

```
using UnityEngine;
using System.Collections;

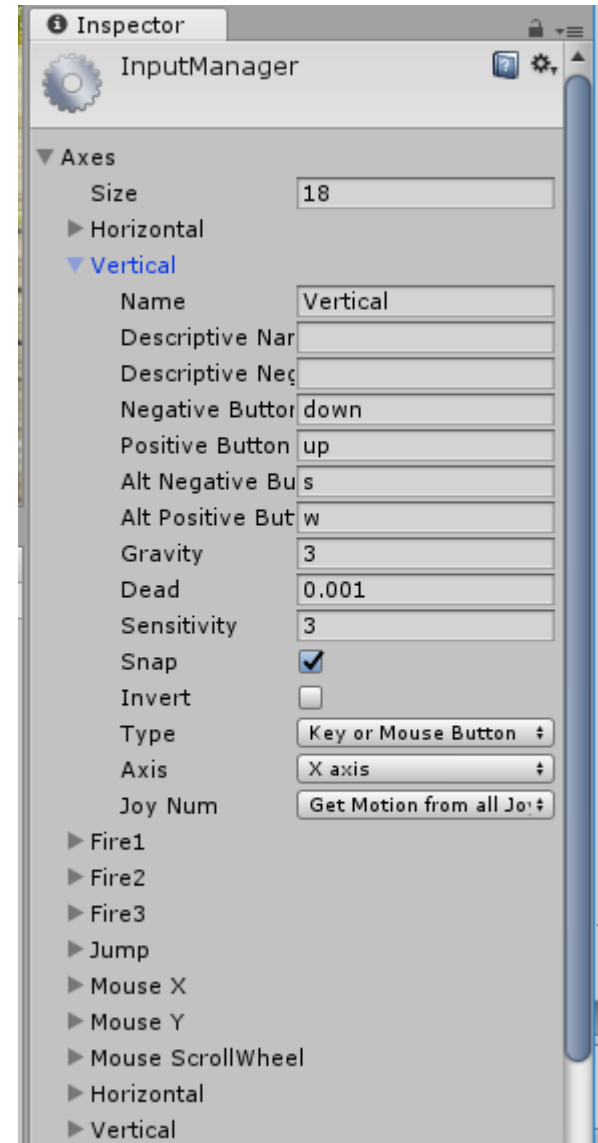
public class ExampleClass : MonoBehaviour {
    void Update() {
        transform.Translate(Vector3.forward * Time.deltaTime);
        transform.Translate(Vector3.up * Time.deltaTime, Space.World);
    }
}
```

- 互動裝置輸入控制
- **GetKey**使用key codes來命名按鍵。
 - 例： `KeyCode.Space` 代表空白鍵
 - 例：判斷按下鍵盤「上」
 - `Input.GetKeyDown(KeyCode. UpArrow);`

定義輸入控制

- GetButton
 - GetButton則是設定好屬於你自己的輸入控制。
 - Edit -> Project Settings -> Input
 - positive為設定的按鈕

`Input.GetButtonDown("Vertical");`
也能判斷是否按下鍵盤「上」

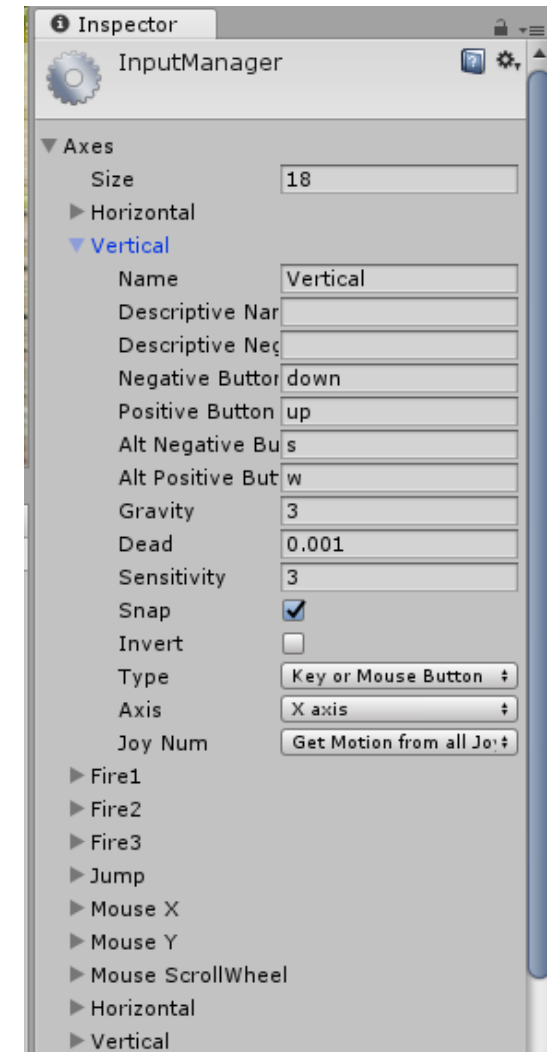


GetButton與GetKey的三種狀態

- GetButtonDown /GetKeyDown：按鈕是否被按下，只有按下時會返回True。
- GetButtonUp/ GetKeyUp：按鈕是否被釋放，只有釋放時會返回True。
- GetButton/GetKey：按鈕是否在按下的狀態，只要按鈕被按下，就會一直傳回True。

搖桿與手把輸入控制

- GetAxis
- GetAxis 會傳回介於-1到1之間的浮點數。
 - Edit -> Project Settings -> Input 中設定參數。
 - 除positive還須設定negative
 - positive往+1移動，negative往-1移動
 - gravity 回到0的速度
 - sensitivity 增加的速度



- GetKey(), GetButton()只需要設定positiv按鈕
- Input.GetAxis設定
 - Positive按鈕
 - negative按鈕
 - gravity, sensitivity, dead 以及snap

▼ Axes	
Size	17
▼ Horizontal	
Name	Horizontal
Descriptive Name	
Descriptive Negative Name	
Negative Button	left
Positive Button	right
Alt Negative Button	a
Alt Positive Button	d
Gravity	3
Dead	0.001
Sensitivity	3
Snap	☒
Invert	☐
Type	Key or Mouse Button
Axis	X axis
Joy Num	Get Motion from all joysticks

當按下**positive**按鈕時，傳回的浮點數會往**+1**慢慢移動。

當按下**negative**按鈕時，傳回的浮點數會往**-1**慢慢移動。

Gravity代表的是放開按鈕時，傳回的浮點數往**0**移動的速度。

Sensitivity代表的是按下按鈕時，傳回的浮點數增加或減少的速度。

Dead 參數越大，則搖桿需要移動越大的距離才能開始返回非零值。

snap選項勾選時，放開原本方向的按鍵，會迅速歸零後再往另一方向移動，而

snap選項未勾選時，則會滑順的往另一方向移動。

```
using UnityEngine;
using System.Collections;

public class ExampleClass : MonoBehaviour {
    void Update() {
        if (Input.GetKey("up"))
            print("up arrow key is held down");

        if (Input.GetKey("down"))
            print("down arrow key is held down");
    }
}
```

```
using UnityEngine;
using System.Collections;

public class ExampleClass : MonoBehaviour {
    public float speed = 10.0F;
    public float rotationSpeed = 100.0F;
    void Update() {
        float translation = Input.GetAxis("Vertical") * speed;
        float rotation = Input.GetAxis("Horizontal") * rotationSpeed;
        translation *= Time.deltaTime;
        rotation *= Time.deltaTime;
        transform.Translate(0, 0, translation);
        transform.Rotate(0, rotation, 0);
    }
}
```

```
using UnityEngine;
using System.Collections;

public class ExampleClass : MonoBehaviour
{
    public GameObject projectile;
    void Update()
    {
        if (Input.GetButtonDown("Fire1"))
            Instantiate(projectile, transform.position, transform.rotation);
    }
}
```

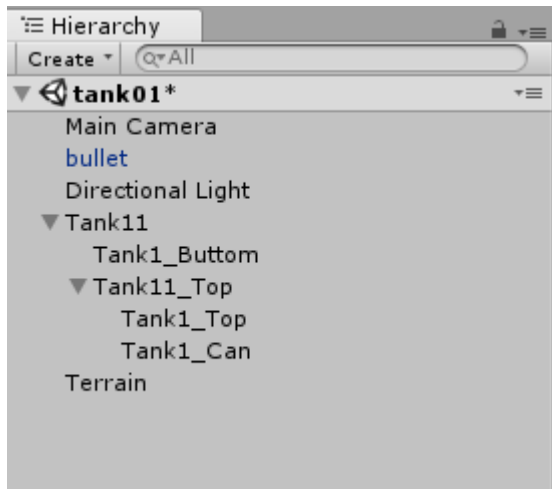
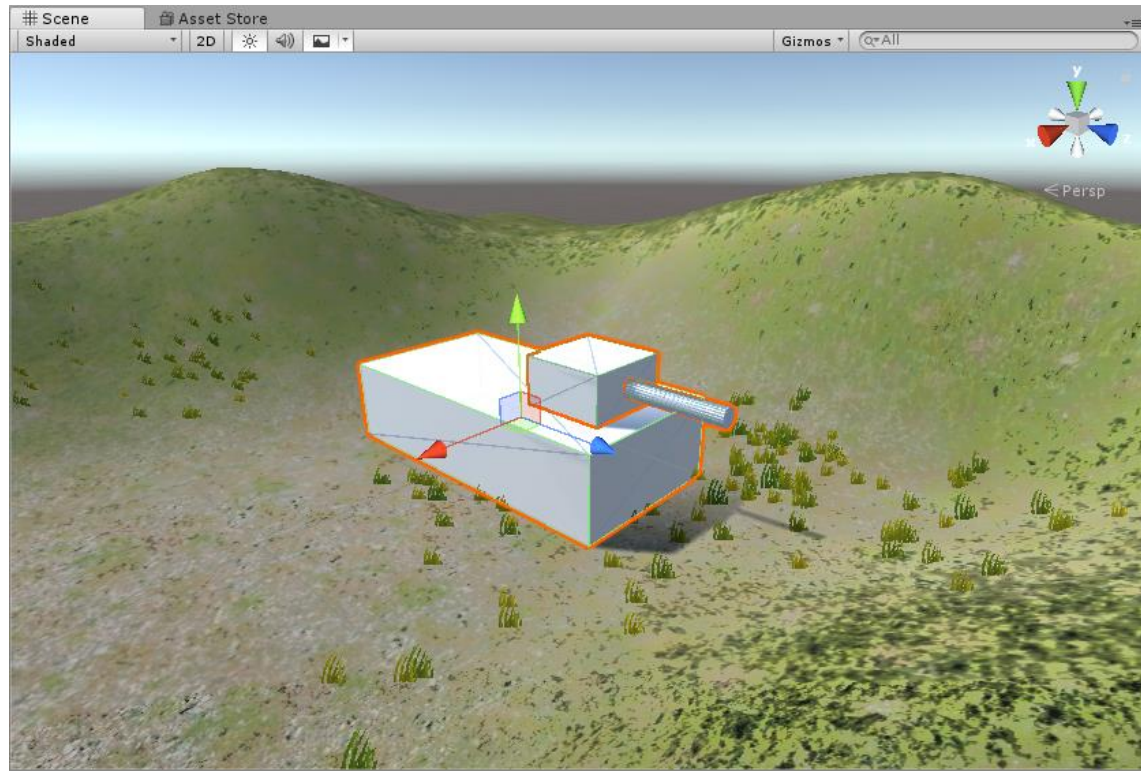
- Example 水平與垂直向量

```
var h = Input.GetAxis ("Horizontal");  
var v = Input.GetAxis ("Vertical");
```

```
transform.Translate (0, 0, 1 *v);  
transform.Rotate (0, 1 *h, 0);
```

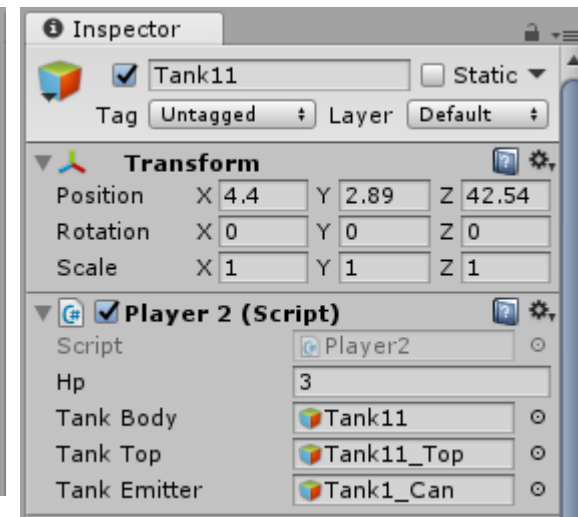
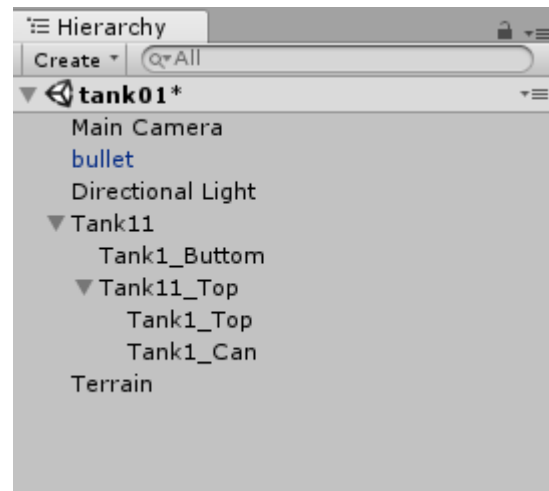
製造坦克車

- 方塊世界坦克車
 - 坦克車底座
 - 坦克車砲塔
 - 坦克車砲管



坦克車移動控制

- 坦克車
 - 前進、後退
 - 左轉彎、右轉彎
- 砲塔
 - 左旋轉、右旋轉
- 砲管
 - 高仰角、低仰角
- 變數指定
 - TankBody: 坦克車本體
 - TankTop: 砲塔
 - TankEmitter: 砲管



```

Player2 ▶ anglenow
1 using System.Collections;
2 using System.Collections.Generic;
3 using UnityEngine;
4
5 public class Player2 : MonoBehaviour {
6     public int Hp = 3;
7     public GameObject TankBody;
8     public GameObject TankTop;
9     public GameObject TankEmitter;
10    int anglenow = 0;
  
```

- 坦克車前進與後退

```
19         if ( Input.GetKey(KeyCode.W) )
20             TankBody.transform.Translate(TankBody.transform.forward*Time.deltaTime*5, Space.World);
21         if ( Input.GetKey(KeyCode.S) )
22             TankBody.transform.Translate(TankBody.transform.forward*Time.deltaTime*-5, Space.World);
```

- 坦克車左轉彎與右轉彎

```
23         if ( Input.GetKey(KeyCode.A) )
24             TankBody.transform.Rotate(new Vector3(0,-1,0), 1);
25         if ( Input.GetKey(KeyCode.D) )
26             TankBody.transform.Rotate(new Vector3(0,1,0), 1);
--
```

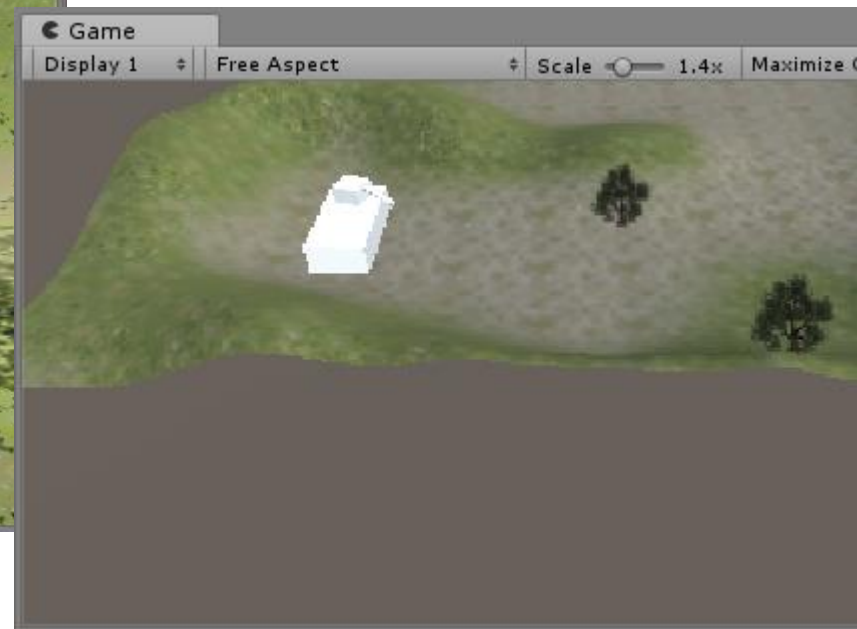
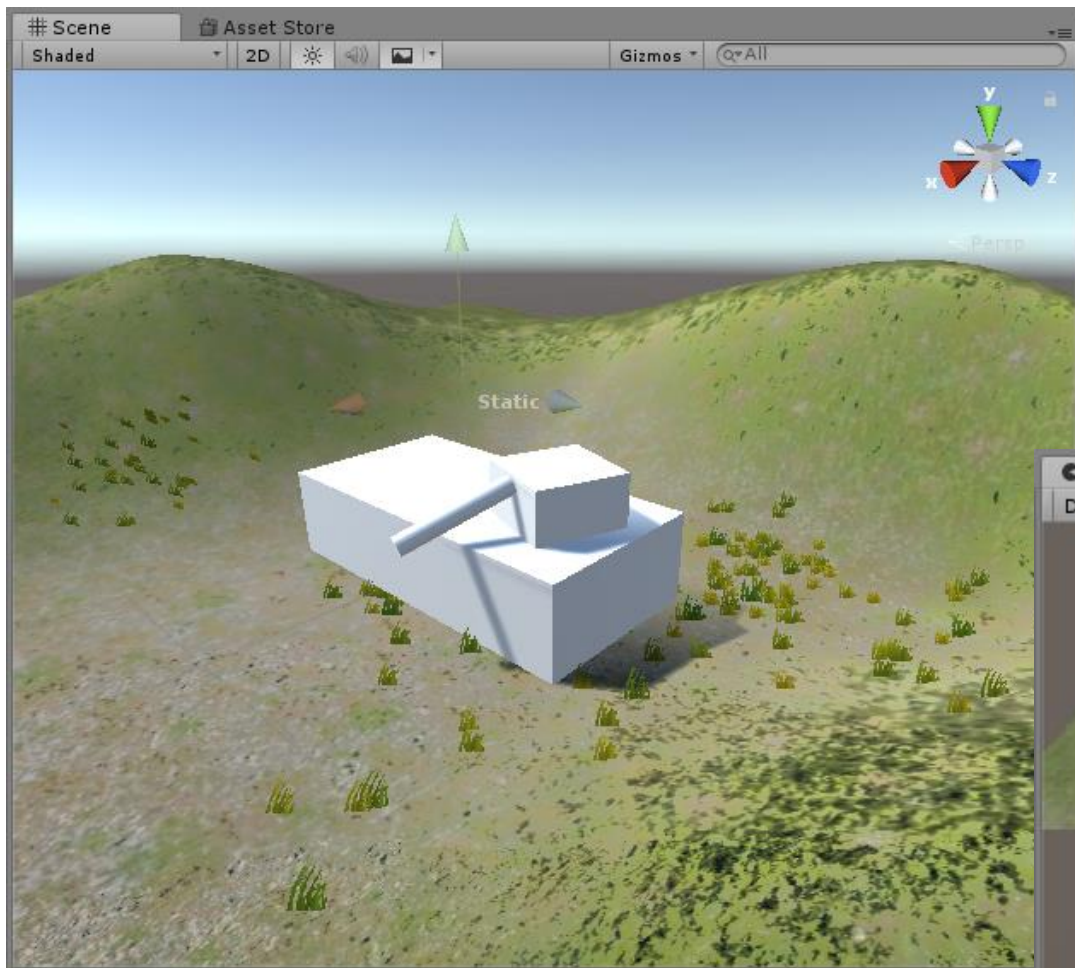
- 砲塔左旋轉與右旋轉

```
28      // Rotate the emitter of tank
29      if ( Input.GetKey(KeyCode.Q) )
30          TankTop.transform.Rotate(new Vector3(0,-1,0), 1);
31      if ( Input.GetKey(KeyCode.E) )
32          TankTop.transform.Rotate(new Vector3(0,1,0), 1);
33      --
```

- 砲管仰角變化

```
34      if ( Input.GetKey(KeyCode.R) )
35      {
36          if( anglenow < 45 )
37          {
38              TankEmitter.transform.Rotate(new Vector3(1,0,0), 1);
39              anglenow++;
40          }
41      }
42      if ( Input.GetKey(KeyCode.F) )
43      {
44          if( anglenow > -10 )
45          {
46              TankEmitter.transform.Rotate(new Vector3(-1,0,0), 1);
47              anglenow--;
48          }
49      }
50      --
```

課堂練習-坦克車的移動控制



Player2 ▶ Update ()

```

1 using System.Collections;
2 using System.Collections.Generic;
3 using UnityEngine;
4
5 public class Player2 : MonoBehaviour {
6     public int Hp = 3;
7     public GameObject TankBody;
8     public GameObject TankTop;
9     public GameObject TankEmitter;
10    int anglenow = 0;
11    // Use this for initialization
12    void Start () {
13
14    }
15
16    // Update is called once per frame
17    void Update () {
18        if (GameMain.currState == GameMain.GameState.Gaming) {
19        if ( Input.GetKey(KeyCode.W) )
20            TankBody.transform.Translate(TankBody.transform.forward*Time.deltaTime*5, Space.World);
21        if ( Input.GetKey(KeyCode.S) )
22            TankBody.transform.Translate(TankBody.transform.forward*Time.deltaTime*-5, Space.World);
23        if ( Input.GetKey(KeyCode.A) )
24            TankBody.transform.Rotate(new Vector3(0,-1,0), 1);
25        if ( Input.GetKey(KeyCode.D) )
26            TankBody.transform.Rotate(new Vector3(0,1,0), 1);
27

```

```

28        // Rotate the emitter of tank
29        if ( Input.GetKey(KeyCode.Q) )
30            TankTop.transform.Rotate(new Vector3(0,-1,0), 1);
31        if ( Input.GetKey(KeyCode.E) )
32            TankTop.transform.Rotate(new Vector3(0,1,0), 1);
33
34        if ( Input.GetKey(KeyCode.R) )
35        {
36            if( anglenow < 45 )
37            {
38                TankEmitter.transform.Rotate(new Vector3(1,0,0), 1);
39                anglenow++;
40            }
41        }
42        if ( Input.GetKey(KeyCode.F) )
43        {
44            if( anglenow > -10 )
45            {
46                TankEmitter.transform.Rotate(new Vector3(-1,0,0), 1);
47                anglenow--;
48            }
49        }
50
51    }
52
53    }
54 }
55

```

坦克車沿著地形高地變化

- 設定物理現象
- Component → Physics
 - Rigidbody
 - Box Collider
- 建立鋼體
- 建立碰撞

